

NEW AI DATA CENTER WILL CONSUME 800,000 GALLONS OF WATER EVERY DAY - IS THIS AMOUNT SOMETHING TO WORRY ABOUT?

800,000 Gallons of water being used everyday sounds like an extraordinarily great amount of water, but is it really that large and worrying of a number? When considering 800,000 gallons of water, we need to ask the question of what 800,000 gallons of water really means in retrospect to many other things before considering whether or not the new Meliora Data Center is a harm to the Rochester community.

The biggest metric for this is what should this be compared to. For the purposes of this writing, let's compare it to our homebase of the University of Rochester. There are a number of metrics to be compared between the data center project and the University of Rochester, but let us focus on the main concern, water. The University used 250 million gallons of domestic water annually and 250 million gallons and sanitary wastewater annually, which should usually not be added together, but for the reasons that 800,000 gallons of water is not specified in the claim, these will be added together for the sake of comparison. The figure that the U of R uses everyday exceeds over 1 million gallons of water given that context.

The community hasn't seemed to raise too many concerns over the water use that the University of Rochester has going on. This is also going along with the benefits that U of R gives back to the community. Alongside the medical center that a lot of the Rochester area is reliant on, the U of R is one of the if not the biggest employer for the Rochester community. Now let's look at the Meliora Data Center. It uses arguably less water while also allowing for the opportunity for jobs, investment,

and opportunities that would benefit the community. Data-centers in particular and the domino effects of lax regulations on such things would engage in more interest in tech investment in Rochester, succeeding the old tech companies that had bubbled here once before. The ROI for the Data-center has potential to significantly help Rochester.

At the same time, there is a cost in negative externalities and impact on the community that may result in the ROI not being worth it.

For example, as mentioned before there is the issue of water and electricity and other costs. We've already established that the water cost isn't much in comparison to other organizations such as the University of Rochester, but it does not change the fact that it adds to the water consumption that Rochester already uses. Additionally, there is an issue that is brought up for the electricity cost of data-centers, which brings about worry of raising costs around the area, thus hurting residents. The exact impact, however, is unknown compared to how much water and electricity that Rochester is already using. What stands true, however, is the fact that there would be more water and electricity usage but there is not a good metric to compare it to.

Essentially, the conclusion of whether the Data-center would prove positive or negative can vary between different evaluations. Compared to the U of R, its impact is not much more and it would appear to act in great benefit to the local area. At the same time, there is a risk of negative externalities which true "cost" or damage cannot be evaluated without looking at the bigger picture, and whether or not that cost is worth it might change depending on how it is being asked.

SOURCES

<https://adminfinance.rochester.edu/departments/energy-services/>
<https://www.rochester.edu/sustainability/waste-diversion-2024/>